Arista Access Node KR-ACC-82 Routing Protocol Flap Recovery SOP

Node: KR-ACC-82 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_4340

Description

In the context of the Arista Access Node KR-ACC-82, a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) scenario occurs due to the dynamic nature of routing protocols, which inherently facilitate network convergence and route advertisement. The access node, being a critical juncture in the network topology, experiences increased protocol activity, particularly in cases where BGP route churn or OSPF adjacency loss is involved. This can lead to an avalanche effect, resulting in ISIS LSP flooding. As a result, the system issues a warning alarm (ALARM_4340) to indicate that the situation is under scrutiny, requiring immediate attention. The root cause analysis will focus on identifying the underlying mechanisms driving this phenomenon and devising a strategic approach to rectify the issue.

Pre-requisite Checks

- Verify that the KR Puram Access site is accessible and secure.

- Check the status of the redundant link to ensure it is operational.

- Gather all relevant information, including alarm details, system logs, and network topology data.

Potential Root Causes

Cause	Probability
BGP route churn caused by frequent network changes or misconfigurations.	High
OSPF adjacency loss due to network congestion, misconfigurations, or hardware issues.	Medium
ISIS LSP flooding resulting from network instability or protocol misconfigurations.	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP summary information, including total number of prefixes, neighbors, and AS number., Description: This command helps identify BGP-related issues by providing a high-level overview of the BGP process, including the number of prefixes, neighbors, and the AS number.

Command: show ospf neighbor, Expected Output: OSPF neighbor information, including neighbor IDs, states, and interface information., Description: This command provides details about OSPF neighbor relationships, including their IDs, states (e.g., Full, Down), and interface information, which can aid in identifying adjacency loss issues.
```

```
Command: show isis database, Expected Output: ISIS database
information, including LSPs, LSDBs, and sequence numbers.,
Description: This command helps analyze the ISIS database, including
LSPs, LSDBs, and sequence numbers, to identify potential flooding
issues.
```

Resolution Steps

Step 1: Step: Stabilize BGP routing through network convergence., Command: clear bgp *, Description: This command resets the BGP process, forcing the system to re-establish BGP sessions and re-advertise routes, which can help stabilize BGP routing and alleviate churn.

Step 2: Step: Re-establish OSPF adjacency., Command: clear ospf process, Description: This command resets the OSPF process, allowing the system to re-establish OSPF adjacency and resolve any adjacency loss issues.

Step 3: Step: Address ISIS LSP flooding., Command: clear isis database, Description: This command resets the ISIS database, helping to alleviate LSP flooding and restore normal ISIS operation.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP summary information
indicating stable operation., Description: Verify that BGP is
operating stably, with no signs of churn or instability.

Command: show ospf neighbor, Expected Output: OSPF neighbor
information indicating healthy relationships., Description: Verify
that OSPF neighbors are stable and operating correctly, with no
adjacency loss issues.

Command: show isis database, Expected Output: ISIS database
information indicating normal operation., Description: Verify that the
ISIS database is stable and operating correctly, with no signs of
flooding.
```

Roll-back Procedures

- Step: Restore previous BGP configuration., Command: copy running-config startup-config, Description: This command saves the current BGP configuration and reverts to the previous configuration, which can help restore normal BGP operation in case of a rollback.

- Step: Re-establish OSPF adjacency., Command: clear ospf process, Description: This command resets the OSPF process, allowing the system to re-establish OSPF adjacency and resolve any adjacency loss issues.

- Step: Address ISIS LSP flooding., Command: clear isis database, Description: This command resets the ISIS database, helping to alleviate LSP flooding and restore normal ISIS operation.